



Ku Yee Siang

AI Agent Engineer

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Languages: Mandarin | Bahasa Melayu | English

EDUCATION

Beijing Institute of Technology | 2022-2026

Bachelor of Science in Artificial Intelligence

GPA: 3.6/4.0 | Best Student Award (2022-2025)

PROJECTS

Harness Agent — Enterprise Procurement Workbench

- Engineered a full-stack procurement agent with Vue 3, FastAPI, DeepSeek, MCP/ERP tools, MongoDB, SQLite, and OpenSandbox for sourcing, inventory analysis, purchase orders, onboarding, and report generation.
- Implemented task planning, specialist sub-agents, SSE execution streaming, human approval for write actions, long-term memory, progressive skill loading, and per-user data and runtime isolation.

Medical LLM Causal Consistency Evaluation Benchmark

- Built MedCausalBenchmark from six NCCN clinical guidelines using a dual-axis taxonomy covering medical scenarios and Basic, Joint, Nested, and Conditional counterfactual reasoning.
- Benchmarked Pure LLM, RAG, and CausalSCM across Qwen, LLaMA, and DeepSeek; CausalSCM achieved 91.39% average accuracy with stronger cross-task consistency.

Mini-Map — Gemini-Powered Virtual Travel Journal

- Developed a choice-driven travel agent with Gemini Pro, Google Cloud Agent Builder, FastAPI, Next.js, and MongoDB Atlas, grounded in real geography and budget.
- Designed validated MCP tools, GeoJSON persistence, Atlas Vector Search with 1024-dimensional embeddings, deduplication, real-world API grounding, and one-click itinerary export.

Agentic Medical Chat System

- Built an end-to-end medical Q&A application combining tool-augmented LLM agents, retrieval, a web interface, and safety guardrails for structured clinical conversations.
- Integrated the frontend, agent orchestration, and backend retrieval workflow into a unified full-stack experience designed for traceable and safer medical information delivery.

TECHNICAL SKILLS

Programming Languages

Python, TypeScript, JavaScript, SQL, HTML/CSS

Agentic AI & LLM Systems

Multi-Agent Orchestration, Agent Task Planning, Tool/Function Calling, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), Prompt & Context Engineering, Human-in-the-Loop Approval, Long-Term Memory

LLM Evaluation & Retrieval

Benchmark Design, Model Comparison, Causal & Counterfactual Evaluation, Embeddings

Backend & APIs

FastAPI, REST APIs, Server-Sent Events (SSE), Authentication & Authorization, API Integration, Input Validation, Error Handling, Report Generation

Frontend Development

Vue 3, React, Next.js, Tailwind CSS, Responsive Web Development